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BAGGAGE HANDLING SYSTEM

This chapter describes the design of Baggage Handling System.

The Baggage Handling System (BHS) Concept Design for Nasiriyah International Airport Project is identified as serving 0.75 MAP (Million Annual Passengers).

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4.1 Executive Summary

BHS Concept Design Basis

The Baggage Handling System (BHS) Concept Design for Nasiriyah International Airport Project is identified as serving 0.75 MAP (Million Annual Passengers).

For Nasiriyah International Airport, the scale of the 0.75 MAP build has been quantified and can, at a high level be summarized as including the following:

- Quantity 1 – 1 Check-In Conveyor with 12 units
- Quantity 1 – Delivery Line to HBS Area
- Quantity 1 – HBS Level 1 and Level 2 Security Screening Line
- Quantity 1 – Make-Up Carousel
- Quantity 1 – Arrivals Reclaim Carousel

BHS Technologies

The BHS will be Designed to handle with the same size and weight characteristics as that of most international systems. Please refer to the table below for specific size and weight requirements:

Size		Length mm	Width mm	Hight mm	Weight kg
Normal Gauge Baggage	Max	900	750	550	50
	Min	250	250	75	2.5

Above size category Out of Gauge (OOG) baggage will be handled at a special counter. Those Super OOG (SOOG) is not conveyable by BHS system.

Size		Length mm	Width mm	Hight mm	Weight kg
OOG	Max	2100	750	550	70
SOOG		> 2100	> 750	> 550	> 70

The BHS System is designed to handle standard baggage size to ensure that all the baggage can be accepted by downstream airports, as well as for inbound Arrivals in Nasiriyah International Airport.

The BHS will be designed based around proven BHS technologies for projects of this scale. At Check-In area, a 3 stage Check-In unit featuring Weigh, Label and Dispatch conveyors is proposed. These will feed through a belt conveyor system which pass through an ECAC standard 3 HBS systems before being transported on to make-up area where bags will be diverted into Inclined Make-Up Carousel.

Based on the BHS scale and layout, the sorting technology adopts manual sorting. All cleared departure bags are conveyed to one Make-Up Carousel, then sorted by handheld scanner. Other sorting technologies typically used in higher capacity Departures BHS's, such as Tilt Tray Sorters (TTS) and Individual Carrier Systems (ICS) are not applicable to this project.

For inbound baggage Horizontal Carousel is proposed as it is optimal for passenger ergonomics.

BHS Design Criteria

Within the overall Basis of Design document, there are several sections that are dedicated to the identification of key BHS design criteria in these include:

- The expected performance of each item of equipment.
- How the BHS should be developed in consideration of correct spacing for maintenance, and operational ergonomics.
- What loads the BHS and EDS equipment will impact upon the building structure.
- Based upon the use of ECAC standard 3 EDS machines, the HBS performance and HBS processes should be followed within the design.
- What are the electrical considerations associated with the design of the BHS
- What international standards will apply to the design of the BHS.

Whilst all of these are detailed in their nature, they provide a very high level of description of how the BHS is planned to be designed to support the 0.75 MAP BHS operations.

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4.2 Definitions & Acronyms

The following Definitions and Acronyms are used in BHS Projects.

Acronyms

The following are the most used acronyms in BHS projects:

- ATR shall mean Automatic Tag Reader
- BCR shall mean Baggage Control Room
- BDD shall mean Baggage Dimensioning Device (aka Volumetric Scanner)
- BHS shall mean Baggage Handling System
- BMS shall mean Building Management System
- BPH / bph shall mean Bags Per Hour
- BPM / bpm shall mean Bags Per Minute or Baggage Processed Message, depending on context
- BPP / bpp shall mean Bags Per Passenger
- BRS shall mean Baggage Reconciliation System
- BSM shall mean Baggage Source Message
- EBS shall mean Early Bag Storage
- ECAC shall mean European Civil Aviation Conference
- EDS shall mean Explosive Detection System
- ETD shall mean Explosive Trace Detection
- E-Stop shall mean Emergency Stop
- FIP shall mean Flights in Process
- GOVT shall mean Guaranteed Operator View Time
- GSE shall mean Ground Service Equipment
- HBS shall mean Hold Baggage Screening
- HBSS shall mean Hold Baggage Screening System
- HLC shall mean High Level Control
- HMI shall mean Human Machine Interface
- HSD shall mean High Speed Diverter
- HVAC shall mean Heating, Ventilating and Air Conditioning
- IATA shall mean International Air Transport Association
- ID shall mean Identification
- IEC shall mean International Electrotechnical Commission
- IED shall mean Improvised Explosive Device
- ISO shall mean International Organization for Standardization

- LAD shall mean Lift Assist Device
- LAN shall mean Local Area Network
- LCD shall mean Liquid Crystal Display
- LED shall mean Light Emitting Diode
- LLC shall mean Low Level Control System
- LM / lm shall mean Linear Meter
- MAP shall mean Million Annual Passengers
- MCT shall mean Minimum Connection Time
- ME shall mean Manual Encoding
- MEC shall mean Manual Encoding Console
- MEP shall mean Mechanical, Electrical and Plumbing System
- MES shall mean Manual Encode Station
- MIS shall mean Management Information System
- O&M shall mean Operations and Maintenance
- OAL shall mean Other Airlines
- OEM shall mean Original Equipment Manufacturer
- OOG shall mean Out of Gauge, synonymous with Over-size and Odd size
- OSR shall mean On-Screen Resolution
- PDP shall mean Power Distribution Point
- PEC shall mean Photo Electric Cell
- PLC shall mean Programmable Logic Controller
- RF shall mean Radio Frequency
- RFID shall mean Radio Frequency Identification
- ROW shall mean Right-of-Way
- SCADA shall mean Supervisory Control and Data Acquisition
- SDIU shall mean Suspect Device Isolation Unit
- STD shall mean Scheduled Time of Departure
- TCU shall mean Threat Containment Unit
- TOB shall mean Top of Belt
- TTS shall mean Tilt Tray Sorter
- UID shall mean Unique Identifier
- ULD shall mean Unit Load Device
- UPS shall mean Uninterrupted Power Supply

- VMU shall mean Vertical Merge Unit
- VSU shall mean Vertical Sort Unit

Definitions

The following are the key definitions used in BHS Projects:

Baggage Handling System: Baggage Handling System, BHS, or System shall mean all components, installation materials, interfaces and other components, all necessary hardware, software, installation coordination and construction supervision of computers/PLC, controls and control hardware and software, management and support services required to implement the work and supply a fully functioning system as described by the Contract Documents.

Sub-system: Sub-system shall mean a part of the BHS, categorized either logically or physically.

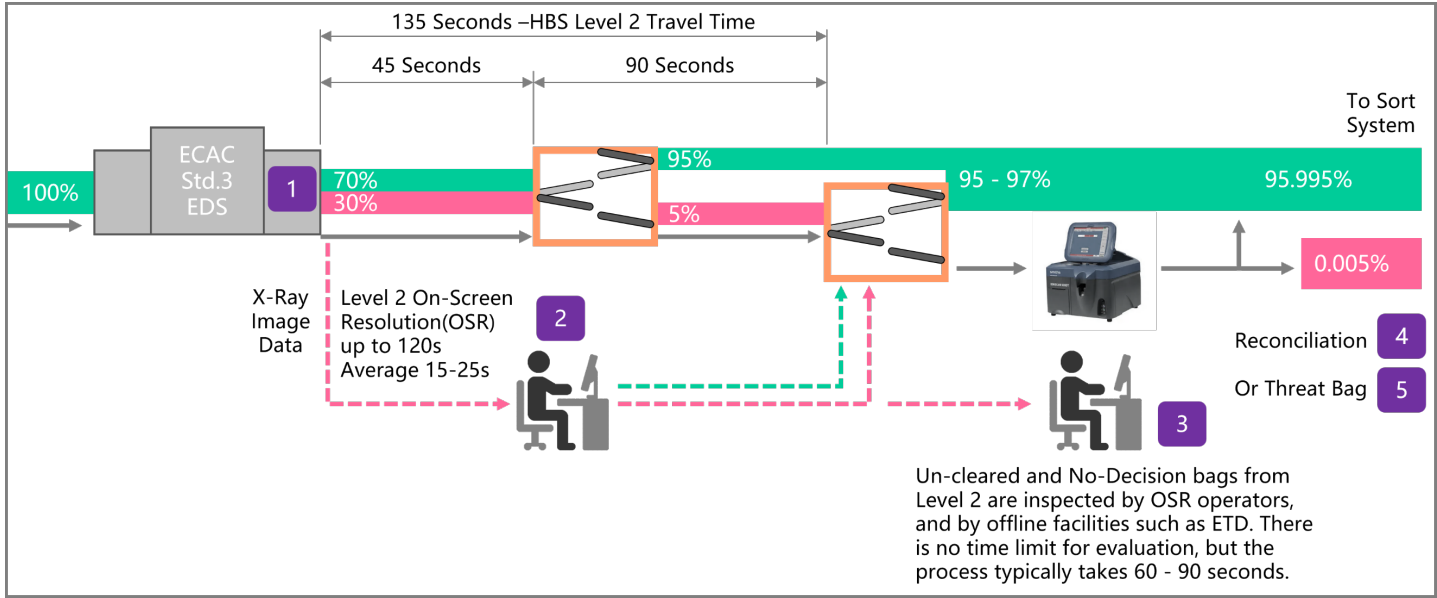
ECAC Standard 3: ECAC has defined a standard to which EDS Machines must conform for use in their HBS applications.

HBS: Hold Baggage Screening is the security screening process identified by a jurisdiction's relevant authority screening of Hold Baggage, i.e., baggage inputted into the BHS at Check-In, Transfer Inputs, Curbside, or other similar Inputs.

HBS features several different levels of escalation of a bag status according to perceived risk. The status of the bag escalates progressively as it fails each increasing level of the process until the point at which it is denoted as cleared or not a threat.

HBS Levels: The following table summarizes the 5 levels associated with the HBS Process proposed for the Nasiriyah International Airport project:

HBS Level	Action	Comment
Level 1	ECAC Standard 3 EDS Machine operating in Automatic Mode	
Level 2A	Operator On-Screen Resolution review of the EDS Machine Image	45 seconds available OSR review time to 1 st decision point
Level 2B	Continued Operator On-Screen Resolution review of the EDS Machine Image	Additional 90 seconds available OSR review time (135 seconds total Review Time across Levels 2A &2B)
Level 3	Operator On-Screen Resolution review of the EDS Machine Image	Operator has an unconstrained image review time, and can make use of offline facilities such as ETD and/or a stand-alone x-ray machine
Level 4	PAX Reconciliation	The Bag is removed from the BHS and taken to the PAX by the security team
Level 5	Threat Bag	Bag is loaded into a Threat Containment Unit, or other explosive mitigating measure



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4.3 BHS Demand Analysis

The Near-term of 0.75 MAP Flight Schedule is received, and the key areas of BHS demand have been calculated, as reported in the following subsections.

Assumptions

To support the BHS Analysis, the key assumptions are used as following.

Assumptions		
Item	Domestic	International
Passenger carrying baggage ratio	80%	100%
Handling Capability per EDS (Primary)	1200 BPH	1200 BPH
Handling Capability per EDS (Secondary)	120 bags	120 bags
EDS Unit Space	100 sqm/EDS	100 sqm/EDS
MUPs per EQA	2	3
Claim Frontage per Passenger	0.45 m/pax	0.5 m/pax
Claim Frontage per Baggage	0.45 m/bag	0.5 m/bag
Ratio of Passenger Collecting Baggage	80%	100%
Peak Occupancy of Passenger	90%	90%
Peak Occupancy of Baggage	90%	90%
Occupy Time by Narrow-Body AC	20 min	20 min
Occupy Time by Wide-Body AC	45 min	45 min

Peak Baggage Demand

The calculation of peak baggage demand is based on the following assumption:

- The percentage of domestic passenger carrying baggage is 80%

- The percentage of international passenger carrying baggage is 100%

Based on the previous calculation on departure peak hour passenger, the peak hour passenger number is 38 for domestic and 250 for international. Therefore, the corresponding peak hour domestic baggage is 40 while international is 250.

Baggage EDS Demand

The calculation of baggage EDS requirements is based on the following assumption:

- The handling capability for EDS is 1200 bags/primary EDS; 120 bags/ secondary EDS
- The secondary EDS throughput rate is 8%
- 100 sqm is needed for each EDS unit

Based on the calculated number of peak hour departure baggage, which is 290 in total, the number of primary EDS is 1. Considering the 8% throughput rate of secondary machine, there’s 1 additional EDS for both domestic and international.

The following table summaries the Departure Baggage EDS Demand.

Baggage EDS Requirements	
Primary EDS Machine	1 unit
Secondary EDS Machine	1 unit

Make-Up Position Demand

The calculation of baggage make-up position is based on the following assumption:

- 2 MUP (Make-Up Position) is required for each domestic EQA (Equivalent Aircraft)
- 3 MUP (Make-Up Position) is required for each international EQA (Equivalent Aircraft)

Combined with the peak hour ATM distribution, which is calculated from flight schedule, there’s only 1 Type B aircraft for domestic flight and 2 Type C aircrafts for international flights. Therefore, 2 MUP is needed for domestic while 3 for international.

Considering 3.6 m/MUP is needed, the total baggage make-up length will be 9 for domestic while 32 for international. Meanwhile, as a standard make-up unit is around 60m and 540 sqm needed for each make-up unit, the total area requirements will be 540 sqm for 1 make-up unit for domestic flights as well as international.

The following table summaries the Departure Baggage Make-Up Demand.

Baggage Make-Up Requirements	
Make-Up Positions	DOM: 2 units / INT: 9 units
Make-Up Length	DOM: 9 m / INT: 32 m
Make-Up Units	1 unit
Make-Up Area	540 sq m

Arrivals Reclaim Demand

Reclaim Demand: The Baggage Arrival Reclaim Demand refers to the baggage claim belt length. The calculation is based on the following assumption:

- The claim frontage is 0.45 m per domestic passenger and baggage as well; 0.5 m per international pax and bag
- The ratio of domestic pax collecting baggage is 80%; 100% for international pax
- The peak occupancy of passenger and baggage is 90% for both domestic and international
- The occupy time by 1 narrow-body aircraft is 20 minutes for domestic and 20 minutes for international

Claim frontage demand is firstly calculated for passenger and baggage respectively, where the number of peak hour arrival baggage and claim frontage per pax/bag is considered. The number of claim frontage per pax/bag is based on IATA LOS (level of service).

Referring to the peak hour aircraft movements which was calculated previously based on flight schedule, the baggage claim units can be calculated considering the occupy time per aircraft.

Since the belt length should satisfy the baggage and waiting passenger, the maximum claim frontage requirement out of pax and bag is taken here. Along with the number of baggage claim units, the total baggage reclaim belt length for domestic arrival is 10 m and 51 m for international arrival.

The following table summaries the Arrival Baggage Reclaim Demand.

Arrival Baggage Reclaim Requirements	
Belt Length for Domestic	10 m
Belt Length for International	51 m

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4.4 Description of BHS Design

Description of Departure BHS

This Part of Ground Floor Plan Layout describes the Departure BHS of the Nasiriyah Project. All descriptions relate to the 0.75 MAP build.

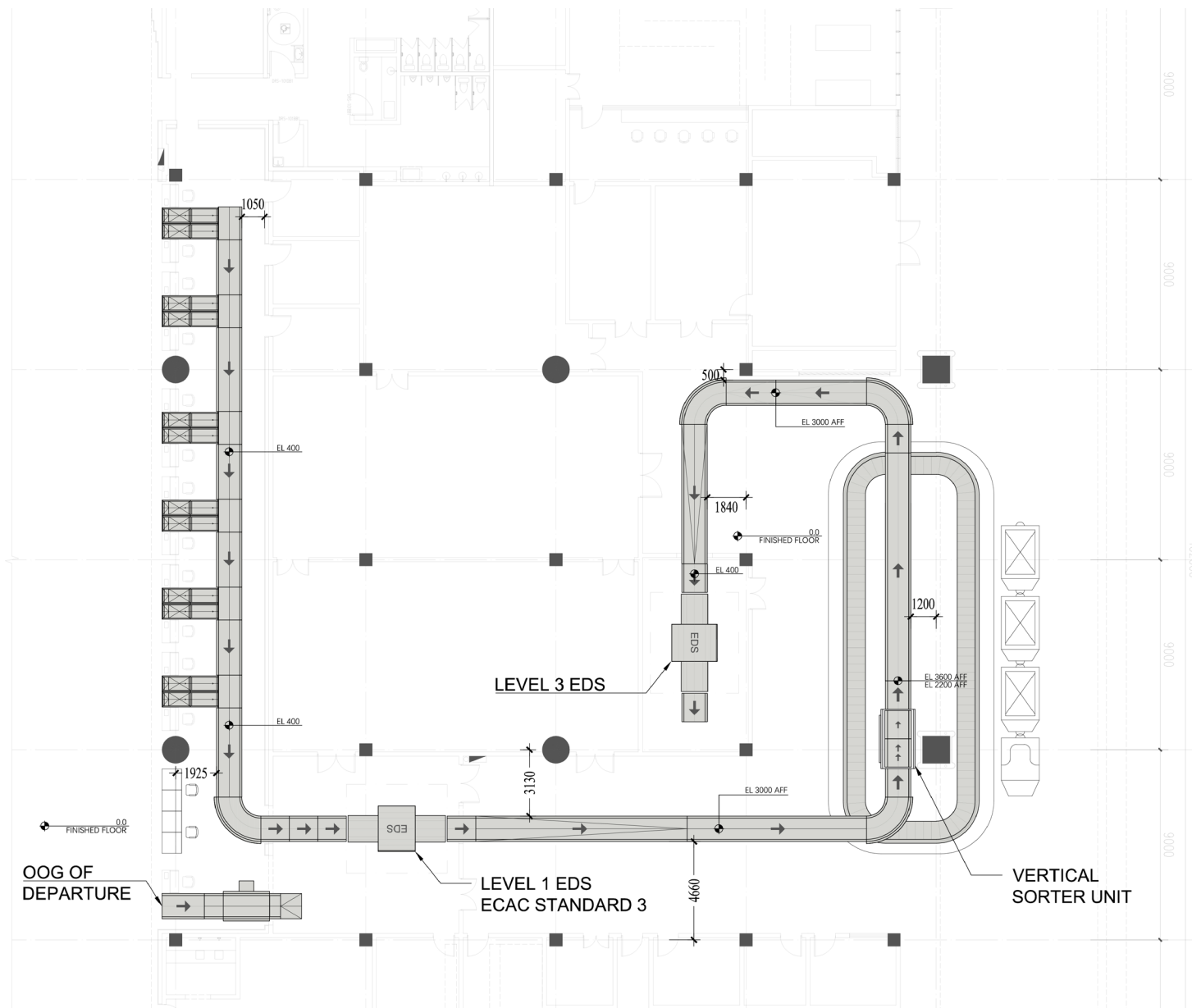
Check-In Level

The Check-In Level BHS features one single ROW of Check-Ins which providing 12 Check-In Positions.

The Check-In Conveyors feature 3-Stage Check-In Conveyors, consisting of the following components:

Weighing/Labelling conveyor: PAX place their baggage onto the first conveyor. An integrated weigh scale weighs the bag. This represents the hand-over point between the PAX and BHS. The check-in agent applies the bag tag at this location.

Dispatch conveyor: The labelled bag is advanced to this conveyor. This conveyor is used to interface with the BHS take-away belt, for automatic discharge into the system. The dispatch conveyor is angled to encourage the bag to fall flat as it transitions from the dispatch Conveyor into the Collector Conveyor.



HBS Operations

HBS Level 1: All bags pass through the EDS machine (referred to as HBS Level 1) without stopping and running at a continuous speed of 0.5m/s, which provides a theoretical throughput rate of slightly less than 30 bpm based on 300mm spacing and an average bag length of around 750mm. However, due to bag conditioning requirements, the BHS cannot consistently feed bags to the EDS machine at such a high rate, so it is for these reasons that the HBS/conveyor line is identified as providing a consistent 20 bpm operational capacity.

Before the bag leaves the EDS machines or in the first few seconds of travel along the conveyor ROW, the EDS machine provides an automated security status decision either clearing the bag or identifying it for further screening. Cleared bags travel along the conveyor ROW until they reach a VSU where they are diverted to a lower level conveyor ROW which is the cleared bag line that routes the bags towards the Make-up Device. From experience

on other projects utilizing HBS Standard 3 Machines, it is expected that the EDS Machine will automatically clear between 70 – 80% of baggage leaving 20-30% uncleared at HBS Level 1, however for the sake of the Level 2 screening workstation demand calculation it is assumed that 30% of bags are rejected at HBS Level 1.

Bags that are not cleared at the HBS Level 1 process are designated as HBS Level 2 bags.

HBS Level 2: The bag leaves the EDS machine and travels at the straight conveyor. This line operates at 0.5m/s which means there is more than 40 seconds for the security operators reviewing the images & 3D model of the bag. If the security operator is satisfied in a timely manner that there is no threat

contained within the bag then the bag is given a clear status by the operator and diverted at the VSU to the cleared line.

It is expected that the Security Operator may clear between 50 – 90% of bags that are referred to the BHS Level 2 process in a timely manner allowing them to be diverted to the cleared line, along with those bags that were cleared at HBS Level 1 by the EDS Machine. In it therefore expected that between 2% and 15% of bags are directed to the Level 3 process.

HBS Level 3: Those bags that are not cleared by the end of the second stage of HBS Level 2 processing are considered to be HBS Level 3.

HBS Level 3 bags are physically routed to a dedicated area where the bag is presented to the operator at one of a number of workstations, where the operator has an unlimited amount of time to review the images from the EDS machine and may also supplement this image review with ETD.

If the operator clears the bag at HBS Level 3 the bag is routed back into the BHS and onto the Make-up Device. Bags that are not cleared at HBS Level 3 may be designated as HBS Level 4 or HBS Level 5.

HBS Level 4: If a Security Operator at HBS Level 3 decides that the bag does not contain an Improvised Explosive Device (IED) but may contain something that still may be a threat to the safety of the aircraft (e.g., dangerous goods), the bag is designated as HBS Level 4. the bag will be removed from the BHS and is manually transported to the passenger for further investigation (or any other process outside of the BHS that the local security services deem necessary). Once outside the BHS a HBS Level 4 bag is not intended to be reintroduced into the BHS and would be manually brought to the make-up area or the aircraft after being cleared by security.

Very few bags each day are designated as HBS Level 4. The exact volumes are impossible to identify as they are very much dependent upon the security processes and protocols of the airport and/or country and the level of threat the city/country is experiencing, the types of bag and types of traveler passing through the airport.

HBS Level 5: If a security operator at HBS Level 3 decides that there is likely to be an IED contained within the bag, then airport protocols for dealing with an IED are implemented. These protocols vary significantly from airport to airport and country to country, so the BHS does not dictate the operation at this point. For example, a bomb disposal team may be dispatched to remove the bag from the Level 3 area directly, the bag could be placed on the Level 4 conveyor and then placed into a Threat Containment Unit (TCU) or there may be other methods to deal with this type of bag as dictated by local security services.

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4.4 Description of BHS Design

Description of Inbound BHS

This Part of Ground Floor Plan Layout describes the Inbound BHS of the Nasiriyah Project. All descriptions relate to the 0.75 MAP build.

As agreed with F&P and the client, the concept includes the use of flat-plate carousel as the claims device in the PAX reclaim hall. This compact device allows for a medley of arrangements owing to its narrow turning radii. The use rubber slats for a surface material are recommended as they are flexible, relatively easy to maintain (and replace), less noisy, and reduce the risk of pinch points for PAX.

The device is well suited in public spaces owing to its ergonomic advantages for PAX claiming bags. Bags easily slide off the surface of the device, and the device is low profiled without lips, or other raised edges, eliminating the need of lifting.

The device can be fed bags either by airside ground handlers who unload directly onto the device, or by an automated delivery system. In this application, the concept shall consider that the bags are offloaded directly onto the device.

Space is reserved outside the baggage carousel for later installation of baggage security equipment.

